

DwC-DP-ALA- Workshop

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Agenda

1. Intro: 3 minutes
2. Data Mapping, Validation, and Publishing: 8 minutes
3. Data indexing and transformation in existing pipelines: 8 minutes
4. DwC-DP Visualization: 8minutes
5. Next steps: 8 minutes
6. Discussion & feedback: 10 minutes



Data Mapping, Validation and Publishing



Darwin Core Data Package (DwC-DP)

Linked CSV tables + structure + dataset metadata

<https://dwc.tdwg.org/dp/>



Mapping and publishing DwC-DP with GBIF IPT

1

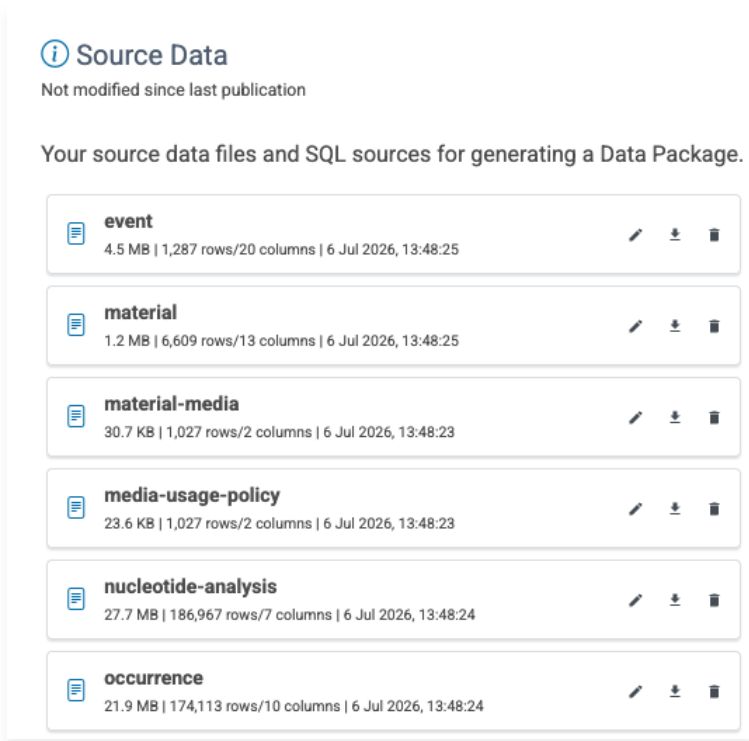
2

3



Upload source data

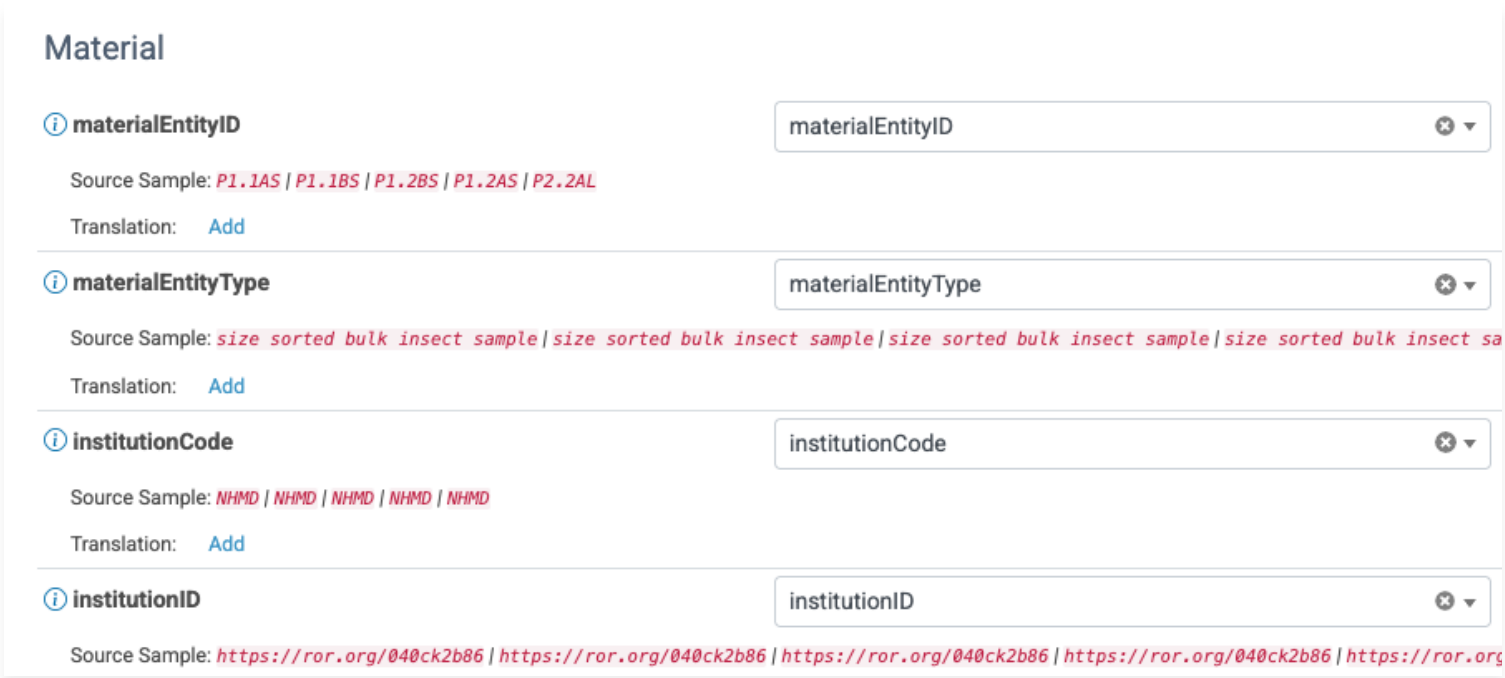
Files, spreadsheets, database



<https://ipt.gbif-test.org/>

Map the data

Multiple DwC-DP schemas



<https://www.gbif.org/ipt>

Fill metadata

Describe the dataset



Mapping and publishing DwC-DP with GBIF IPT

4

Validate
Check and resolve issues

Data Package Validation Report

Metadata validation

2 issues found. See details below.

Severity	Type	Message	Location
WARNING	MISSING_NAME	The 'name' property is missing or blank.	/dataset/title
ERROR	UNRECOGNIZED_PROFILE_VERSION	Unrecognized DwC-DP profile version 'data-package' - JSON Schema and table schema validation skipped.	/profile

EML validation

3 issues found. See details below.

Severity	Type	Message	Location
WARNING	EML_MISSING_TITLE	EML file is missing a required <title> element or it is empty.	/dataset/title
WARNING	EML_MISSING_CREATOR	EML file is missing a required <creator> element or it is empty.	/dataset/creator
WARNING	EML_XSD_VIOLATION	XSD error at line 8: cvc-complex-type 2.4.a: Invalid content was found starting with element 'pubDate'. One of {alternateIdentifier, shortName, title, references} is expected.	

Data validation

TABLES

14

TOTAL ROWS

618,555

TABLES WITH NO ISSUES

6 / 14

INTEGRITY ISSUES

9

material 6,609 rows 1 issue(s)

Primary key violations found.

Field(s)	Violations	Sample values
materialEntity_pk	3	count_star()>2, materialEntity_pk#P112:185 count_star()>2, materialEntity_pk#P112:185_raw count_star()>2, materialEntity_pk#P112:185_pure

Column	Populated	Unique values	Note
materialEntity_pk	100%	6,606	
materialEntityID	100%	6,606	
materialEntityCategory	100%	3	
materialEntityType	100%	2	
institutionCode	100%	1	
institutionID	100%	1	
ownerInstitutionCode	100%	1	
ownerInstitutionID	100%	1	
collectionCode	33%	1	
catalogNumber	33%	1	
preparations	100%	3	
disposition	100%	2	
derivedFromMaterialEntityID	67%	4,404	
derivationEvent_fk	100%	1,287	

5

Publish
Release the DwC-DP dataset

Publication

Edit

Publish

Options

Publishing Organization: Test Organization #1

A preview of your pending published version compared with the current version if existing.

Version 3

Published on 13 Jul 2026, 13:50:43

CURRENTCC0 1.0

Version 4

Next publication date not set

PENDINGCC0 1.0

https://ipt.gbif-test.org/

Mapping and publishing DwC-DP Resources and repos

- GBIF Test IPT: <https://dwcdp-ipt.gbif-test.org/>
 - Validation report: https://dwcdp-ipt.gbif-test.org/manage/validationReport.do?r=insektmobilen_celie_test
- DwC-DP Mapper Prototype: <http://130.225.43.35:3010/>
- DwC-DP Analyser: <https://github.com/gbif/dwc-dp-analyser-service>
- DwC-DP Repository: <https://github.com/gbif/dwc-dp>



• Mapping and publishing DwC-DP GBIF API additions

• Dataset/Registry API

- EndPoint Type: https://api.gbif-test.org/v1/dataset/search?ENDPOINT_TYPE=DWC_DP&facet=DWC_DP_RESOURCE_SCHEMA
- Resource Schema: https://api.gbif-test.org/v1/dataset/search?DWC_DP_RESOURCE_SCHEMA=table-schemas/identification.json
- Validation JSON Report: <https://api.gbif-test.org/v1/dataset/cb11518b-4e42-4697-85f2-a48228cec0e4/validationreport>

• Datapackage

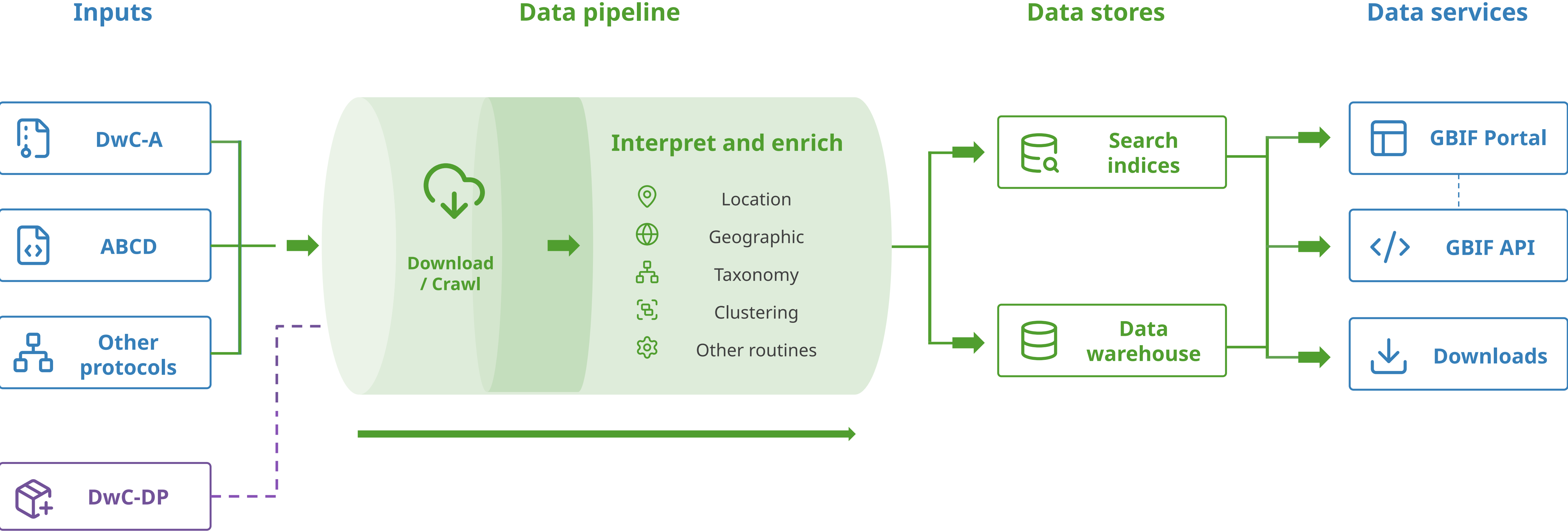
- List: <https://api.gbif-test.org/v1/dataset/datapackage/>
- Detail (datapackage.json): <https://api.gbif-test.org/v1/dataset/cb11518b-4e42-4697-85f2-a48228cec0e4/datapackage>
- Tables/resources: <https://api.gbif-test.org/v1/dataset/cb11518b-4e42-4697-85f2-a48228cec0e4/datapackage/resource>
- Resource/table detail: <https://api.gbif-test.org/v1/dataset/cb11518b-4e42-4697-85f2-a48228cec0e4/datapackage/resource/agent>
- Resource Names: <https://api.gbif-test.org/v1/dataset/cb11518b-4e42-4697-85f2-a48228cec0e4/datapackage/resourceNames>



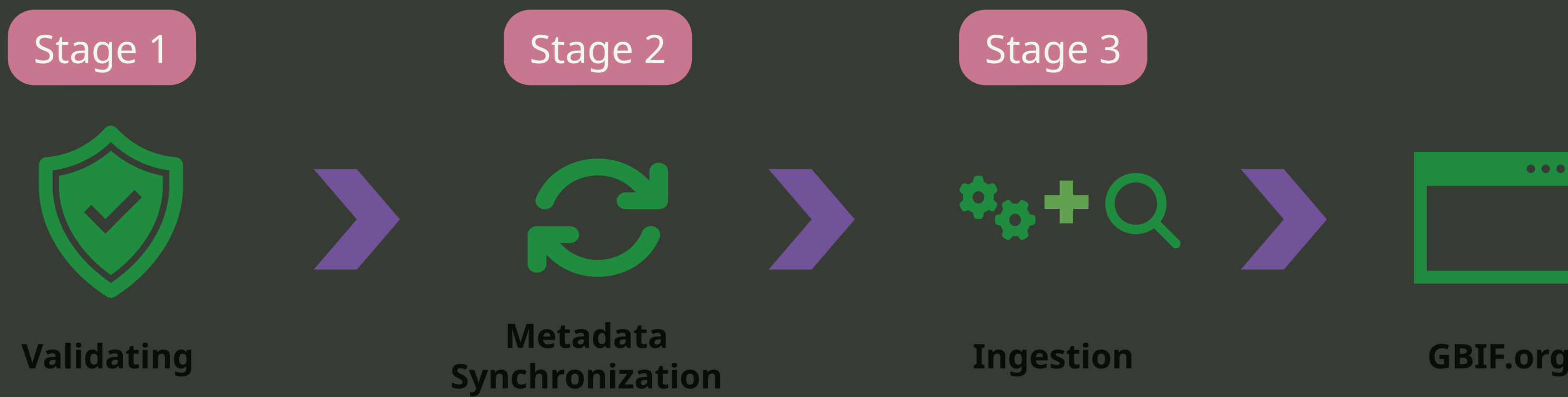
Data indexing and transformation in existing pipelines



GBIF data ingestion pipeline



Integrating DwC-DP



Darwin Core Data Package



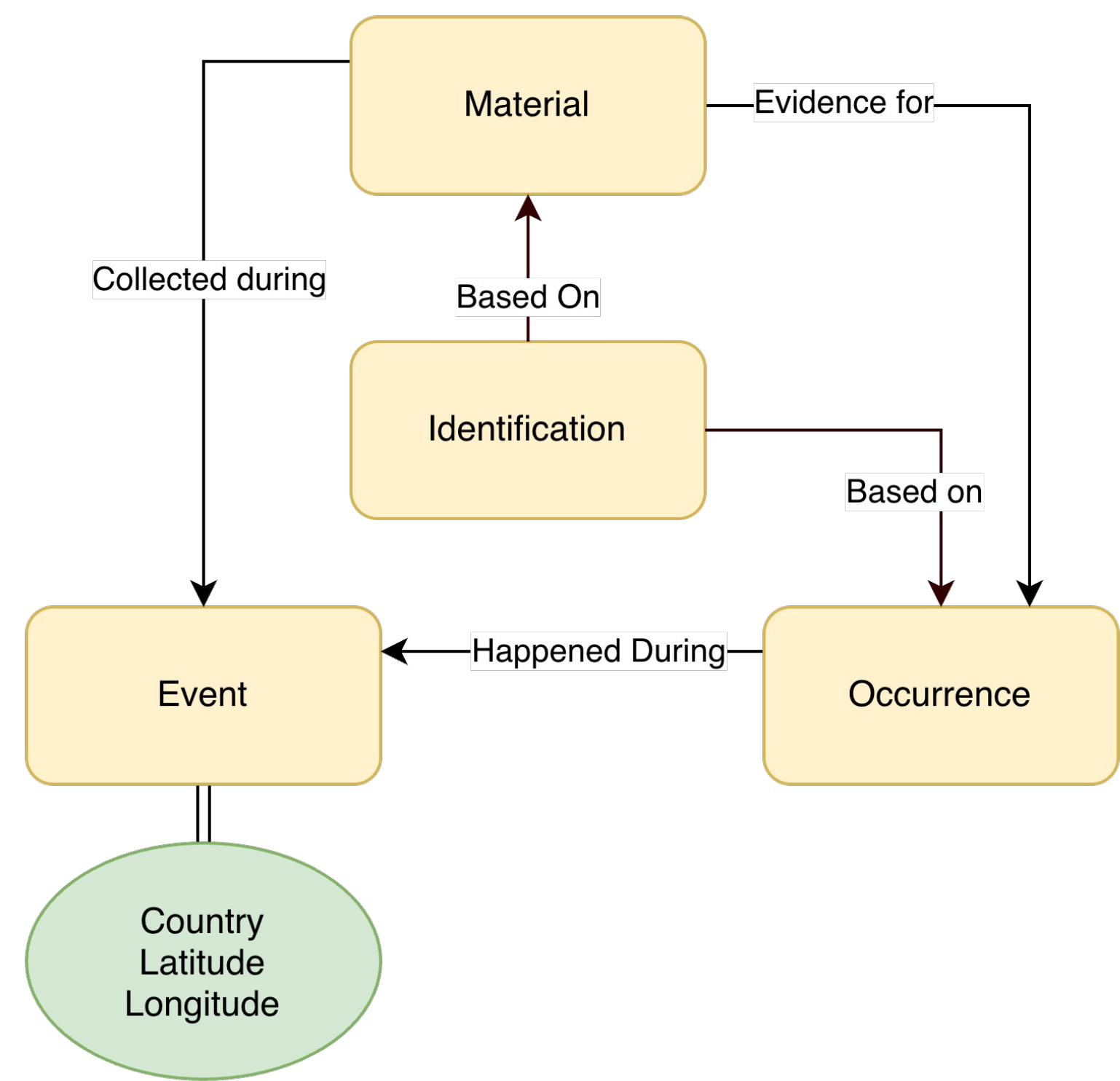
Ingestion and Transformation

Datase
t
Basis of
record

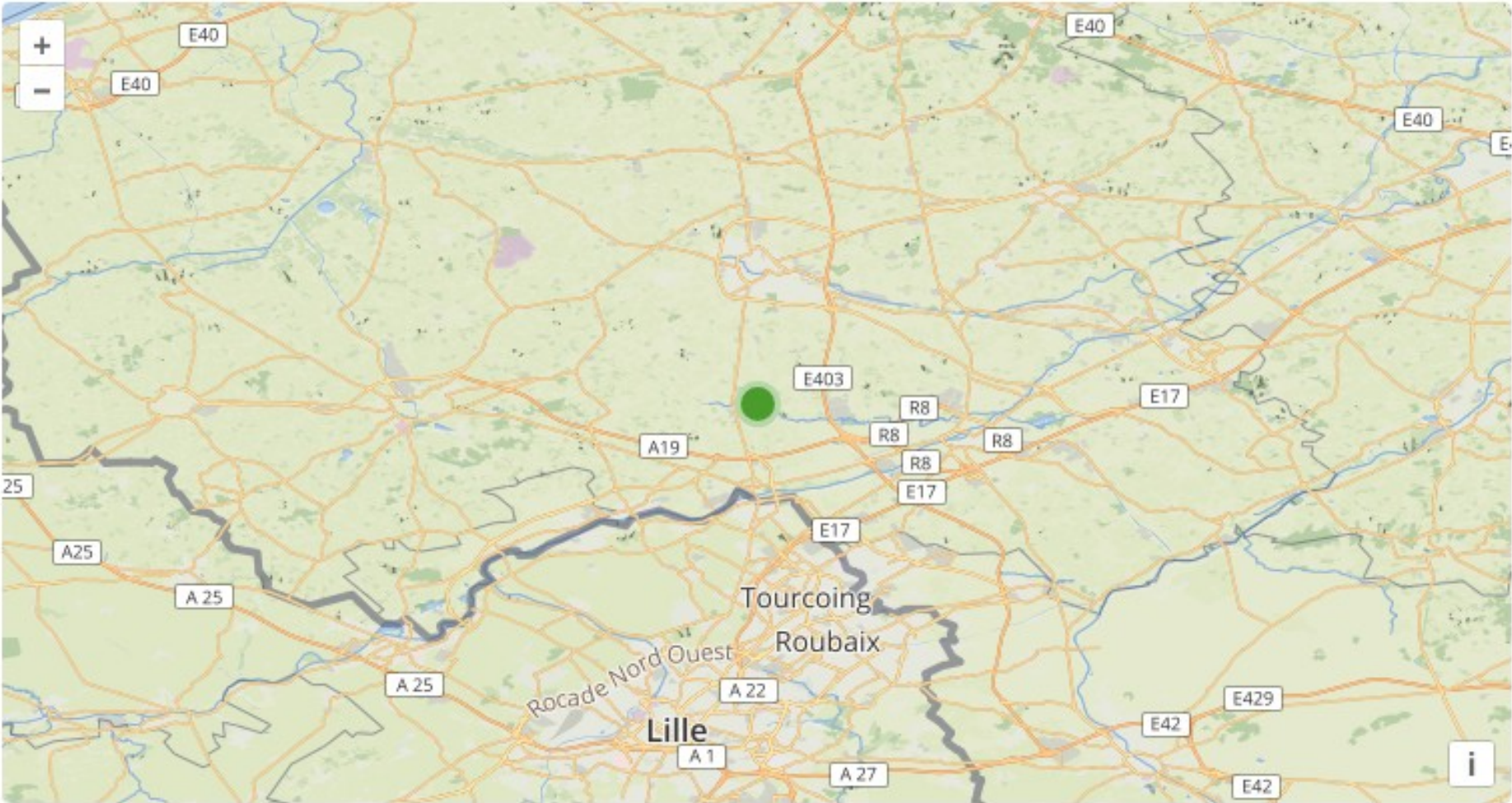
Meise Botanic Garden Herbarium
(BR)
Occurrenc
e



Geographical Mapping



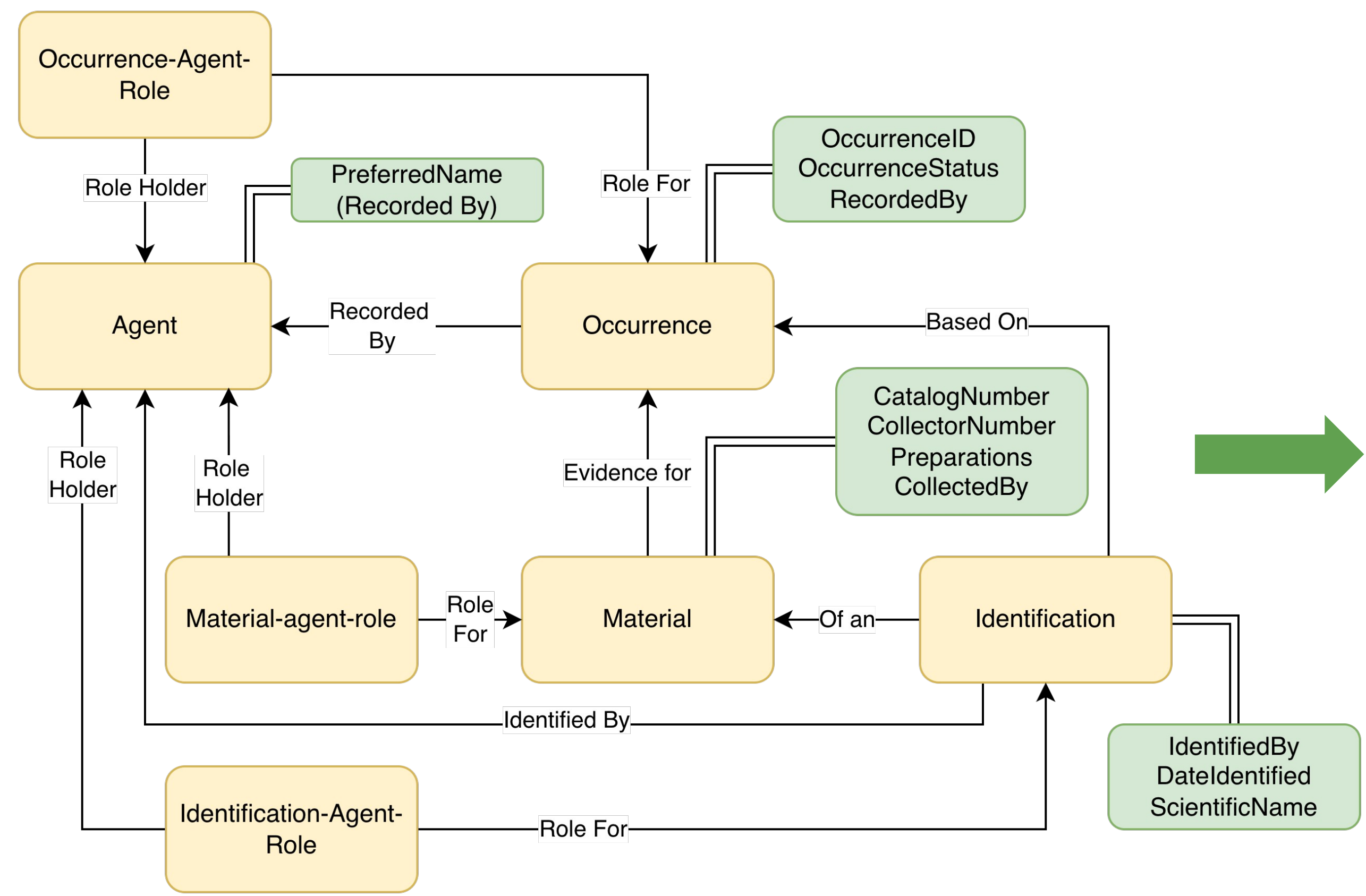
Location



Continent	Europe Inferred
Country or area	Belgium
Country code	BE
Locality	Beitem towards Ledegem, former railway track
Decimal latitude	50.854444
Decimal longitude	3.125833
Location according to GADM	Belgium > Vlaanderen > West-Vlaanderen > Roeselare



Material Mapping

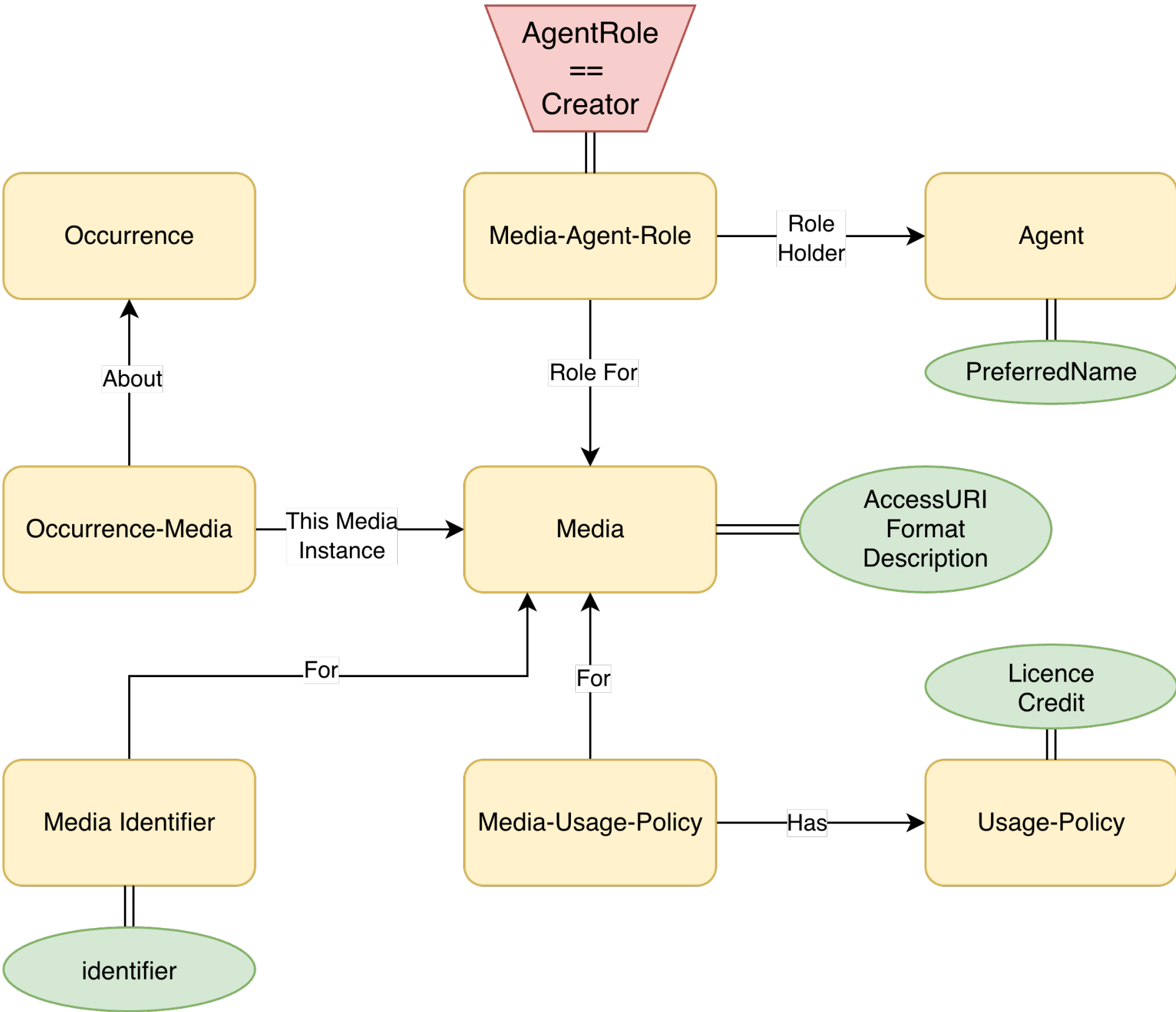


Occurrence

Occurrence ID	http://www.botanicalcollections.be/specimen/BR0000026502854V
Catalogue number	BR0000026502854V
Record number	14631
Recorded by	Verloove F.
Recorded by ID	Filip Verloove https://orcid.org/0000-0003-4144-2422
Occurrence status	Present present original
Preparations	HerbariumSheet



Media Mapping



Description	Image of herbarium specimen BR0000026502854V
Format	image/jpeg
Identifier	https:// oxalis.br...000026502854V.jpg
Creator	Meise Botanic Garden
Record licence	CC BY 4.0
Suggested attribution	"Poa infirma Kunth" - <i>Poa infirma</i> Kunth recorded in Belgium (licensed under CC BY 4.0)



Living Atlases

- Re-integrate pipelines
- Sensitive species?
- New data services?



DwC-DP Visualization



New views in GBIF.org - Validation and metrics report



- Viewable by publishers to easily get insight into their specific dataset
- Consistent with the IPT
- Example:
 - Dataset page: <https://www.gbif-test.org/dataset/fdd19466-79c8-4247-a7d0-6651b640158e>



New views in GBIF.

Navigation

humboldt-survey

sampling event · 10.21373/krfb9y · 4a2d5856-250e-4536-9ef6-e9a020203eaf

Tables (6)		
	ROW COUNT	RELATIONS
CORE 5		
event	2,500	
survey	1,800	
occurrence	32,540	
survey-target	40	
organism	4,000	
RELATION ONLY 1		
survey-survey-target	2,695	

Click a table to see its columns.

Insektnmobilen - National citizen science and DNA metabarcoding survey of flying insects in June 2018 and 2019

sampling event · 10.21373/cuzgvu · ab0aae97-5fa4-4205-99ae-ab1379bf2af1

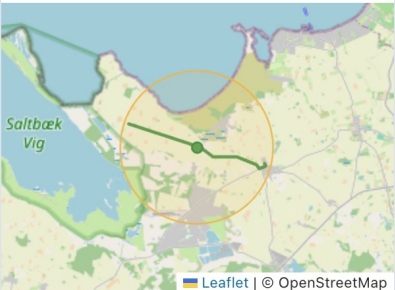
Crawl attempt 34 · 2026-09-14 · 49

Tables > event > P10.1BS	
Record 4 of 8	
within eventCategory is nucleotide analysis eventConductedByID is P10	
FIELD	VALUE
event_pk	P10.1BS
3 materials 252 nucleotide-analysis 259 occurrences 1 survey	
parentEvent_fk	happened during P10.1B
eventCategory	nucleotide analysis
datasetName	Insketmobilen
eventConductedByID	conducted by P10
eventDate	2018-06-24 17:45:00/2018-06-24 18:10:00
habitat	land cover: agriculture subland cover: intensive agri...
	Cloudless
	5674
	Denmark
	DK
	Eskebjerg
	55.71945068
	11.26753341
	2500
	MULTILINESTRING ((11.232379999227009 55.726359999866062...
	footprint
	collector identities withheld

LOCATION

event.decimallatitude / footprintWKT

the point, its stated uncertainty, and the footprint



55.71945068, 11.26753341 ± 2,500 m · footprint: multilinestring

Expand

Insektnmobilen - National citizen science and DNA metabarcoding survey of flying insects in June 2018 and 2019

sampling event · 10.21373/cuzgvu · ab0aae97-5fa4-4205-99ae-ab1379bf2af1

Crawl attempt 34 · 2026-09-14 · 49

Tables > event		Columns	Rows
2,202 rows · 18 columns · primary key: event_pk · id: eventID			
+ Filter			
FIELD		POPULATED	UNIQUE
event_pk	string NN UQ	100%	2,202
event-media material nucleotide-analysis occurrence survey event (self)			
parentEvent_fk	string	100%	1,287
happened during event (self)			
eventCategory	string NN	100%	1
datasetName	string	100%	1
eventConductedByID	string	100%	345
conducted by agent			
eventDate	string	100%	1,266
habitat	string	100%	50
eventRemarks	string	19%	214
locationID	string	99%	439
country	string	100%	1
countryCode	string	100%	1
locality	string	100%	539
decimallatitude	number	100%	640
decimallongitude	number	100%	661
coordinateUncertaintyInMeters	number	100%	1
footprintWKT	string	95%	470
preferredSpatialRepresentation	string	100%	1
informationWithheld	string	100%	1

Insektnmobilen - National citizen science and DNA metabarcoding survey of flying insects in June 2018 and 2019

sampling event · 10.21373/cuzgvu · ab0aae97-5fa4-4205-99ae-ab1379bf2af1

Crawl attempt 34 · 2026-09-14 · 49

Tables > event

Columns

Rows

<1-8 of 8>

eventCategory is nucleotide analysis

eventConductedByID is P10

+ Filter

Clear all

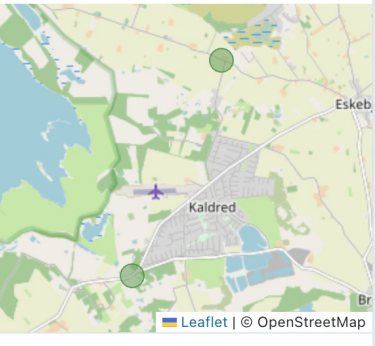
#	event_pk	parentEvent_fk	eventCategory	datasetName	eventConductedByID	eventDate
1	P10.1AL	P10.1A	nucleotide analysis	Insketmobilen	P10	2018-06-
2	P10.1AS	P10.1A	nucleotide analysis	Insketmobilen	P10	2018-06-

n science and DNA metabarcoding survey of 2019

LOCATIONS

event.decimallatitude

where the filtered rows are



8 of 2,202 mapped · 2 places

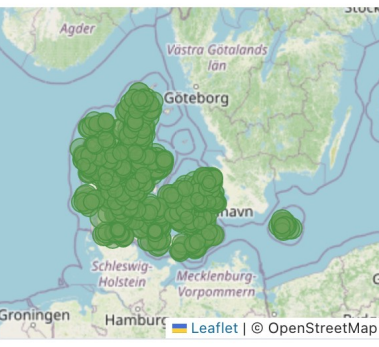
Expand

Pick a column's ID to see how its values are distributed, and it to keep it.

LOCATIONS

event.decimallatitude

where the rows are, from the real coordinates



2,202 of 2,202 mapped · 663 places

Expand

Pick a column's ID to see how its values are distributed, and it to keep it.



New views in GBIF.org - DwC-DP navigation

Table Detail



Insektmobilen - National citizen science and DNA metabarcoding survey of flying insects in June 2018 and 2019

sampling event · 10.21373/cuzgvu · ab0aae97-5fa4-4205-99ae-ab1379bf2af1

Crawl attempt 34 · 2026-09-14 · 49

Tables > event

Columns Rows

2,202 rows · 18 columns · primary key: event_pk · id: eventID

+ Filter

FIELD	POPULATED	UNIQUE
event_pk string NN UQ	100%	2,202
event-media		
material		
nucleotide-analysis		
occurrence		
survey		
event (self)		
parentEvent_fk string	100%	1,287
happened during event (self)		
eventCategory string NN	100%	1
datasetName string	100%	1
eventConductedByID string	100%	345
conducted by agent		
eventDate string	100%	1,266
habitat string	100%	50
eventRemarks string	19%	214
locationID string	99%	439
country string	100%	1
countryCode string	100%	1
locality string	100%	539
decimallatitude number	100%	640
decimallongitude number	100%	661
coordinateUncertaintyInMeters number	100%	1
footprintWKT string	95%	470
preferredSpatialRepresentation string	100%	1
informationWithheld string	100%	1

LOCATIONS

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Pick a column's to see how its values are distributed, and it to keep it.

Insektmobilen - National citizen science and DNA metabarcoding survey of flying insects in June 2018 and 2019

sampling event · 10.21373/cuzgvu · ab0aae97-5fa4-4205-99ae-ab1379bf2af1

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252 nucleotide-analysis	
259 occurrences	
1 survey	
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eventCategory	nucleotide analysis
datasetName	Insektmobilen
eventConductedByID	conducted by P10
eventDate	2018-06-24 17:45:00/2018-06-24 18:10:00
habitat	land cover: agriculture subland cover: intensive agri...
eventRemarks	Cloudless
locationID	5674
country	Denmark
countryCode	DK
locality	Eskebjerg
decimallatitude	55.71945068
decimallongitude	11.26753341
coordinateUncertaintyInMeters	2500
footprintWKT	MULTILINESTRING ((11.232379999227009 55.726359999866062...
preferredSpatialRepresentation	footprint
informationWithheld	collector identities withheld

LOCATION

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sampling event · 10.21373/cuzgvu · ab0aae97-5fa4-4205-99ae-ab1379bf2af1

Crawl attempt 34 · 2026-09-14 · 49

Tables > event

Columns Rows

1-8 of 8

eventCategory is nucleotide analysis eventConductedByID is P10

+ Filter Clear all

#	event_pk	parentEvent_fk	eventCategory	datasetName	eventConductedByID	eventDate
1	P10.1AL	P10.1A	nucleotide analysis	Insektmobilen	P10	2018-06-24 17:45:00
2	P10.1AS	P10.1A	nucleotide analysis	Insektmobilen	P10	2018-06-24 17:45:00
3	P10.1BL	P10.1B	nucleotide analysis	Insektmobilen	P10	2018-06-24 17:45:00
4	P10.1BS	P10.1B	nucleotide analysis	Insektmobilen	P10	2018-06-24 17:45:00
5	P10.2AL	P10.2A	nucleotide analysis	Insektmobilen	P10	2018-06-24 18:10:00
6	P10.2AS	P10.2A	nucleotide analysis	Insektmobilen	P10	2018-06-24 18:10:00
7	P10.2BL	P10.2B	nucleotide analysis	Insektmobilen	P10	2018-06-24 18:10:00
8	P10.2BS	P10.2B	nucleotide analysis	Insektmobilen	P10	2018-06-24 18:10:00

LOCATIONS

event.decimallatitude

where the filtered rows are

8 of 2,202 mapped · 2 places

Expand

Pick a column's to see how its values are distributed, and it to keep it.

Mapping and publishing DwC-DP with GBIF IPT

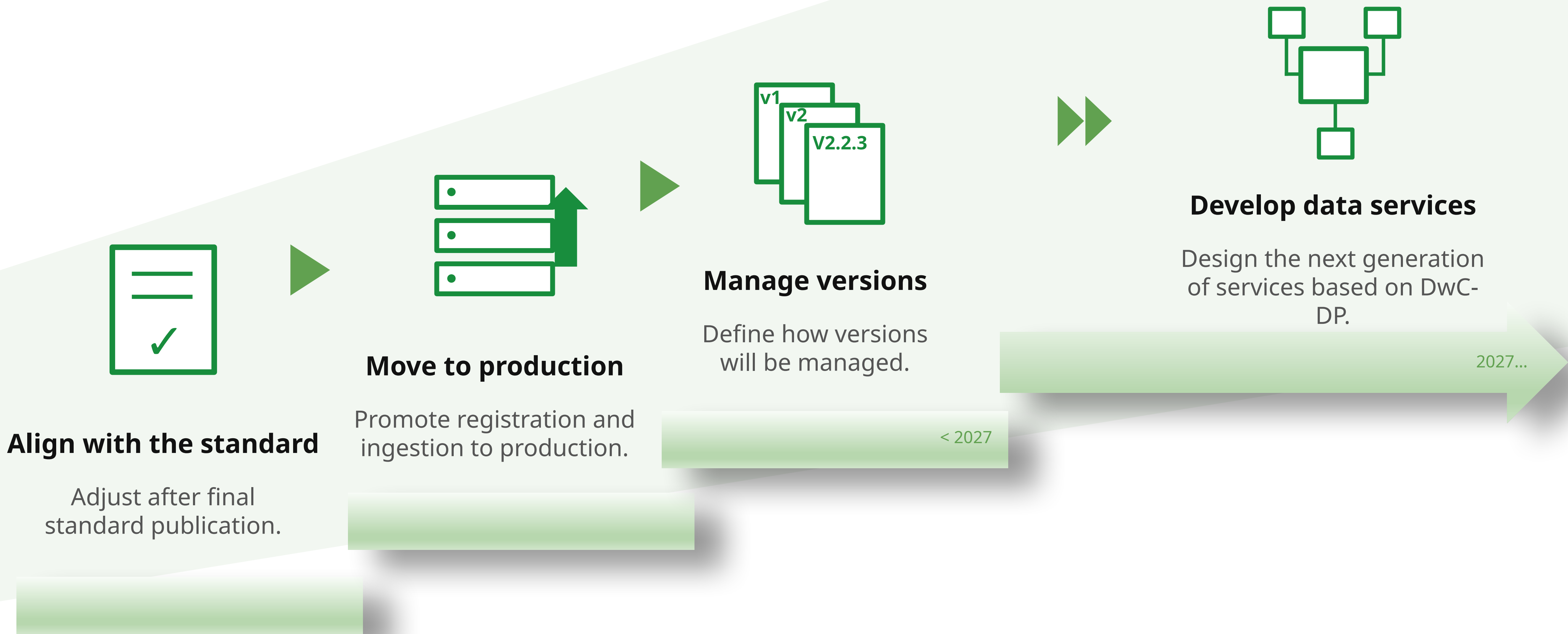
- DwC-DP Explorer:
 - v0.001: <http://130.225.43.35:3000/>
 - V0.9: <http://dwcp-explorer.labs.gbif.org/>
- GeoParquet Example: https://labs.gbif.org/~mblissett/pt_dash.html



Next Steps

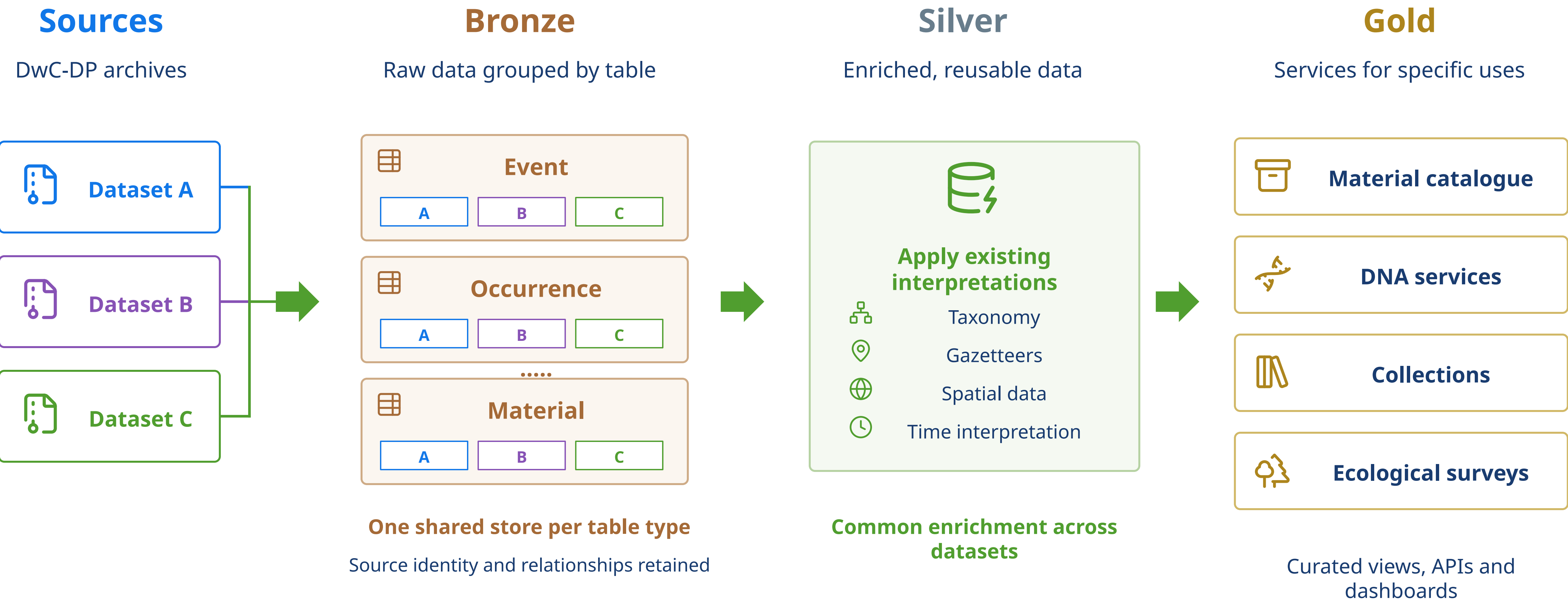


Future plans for DwC-DP in GBIF



DwC-DP in a Medallion Architecture

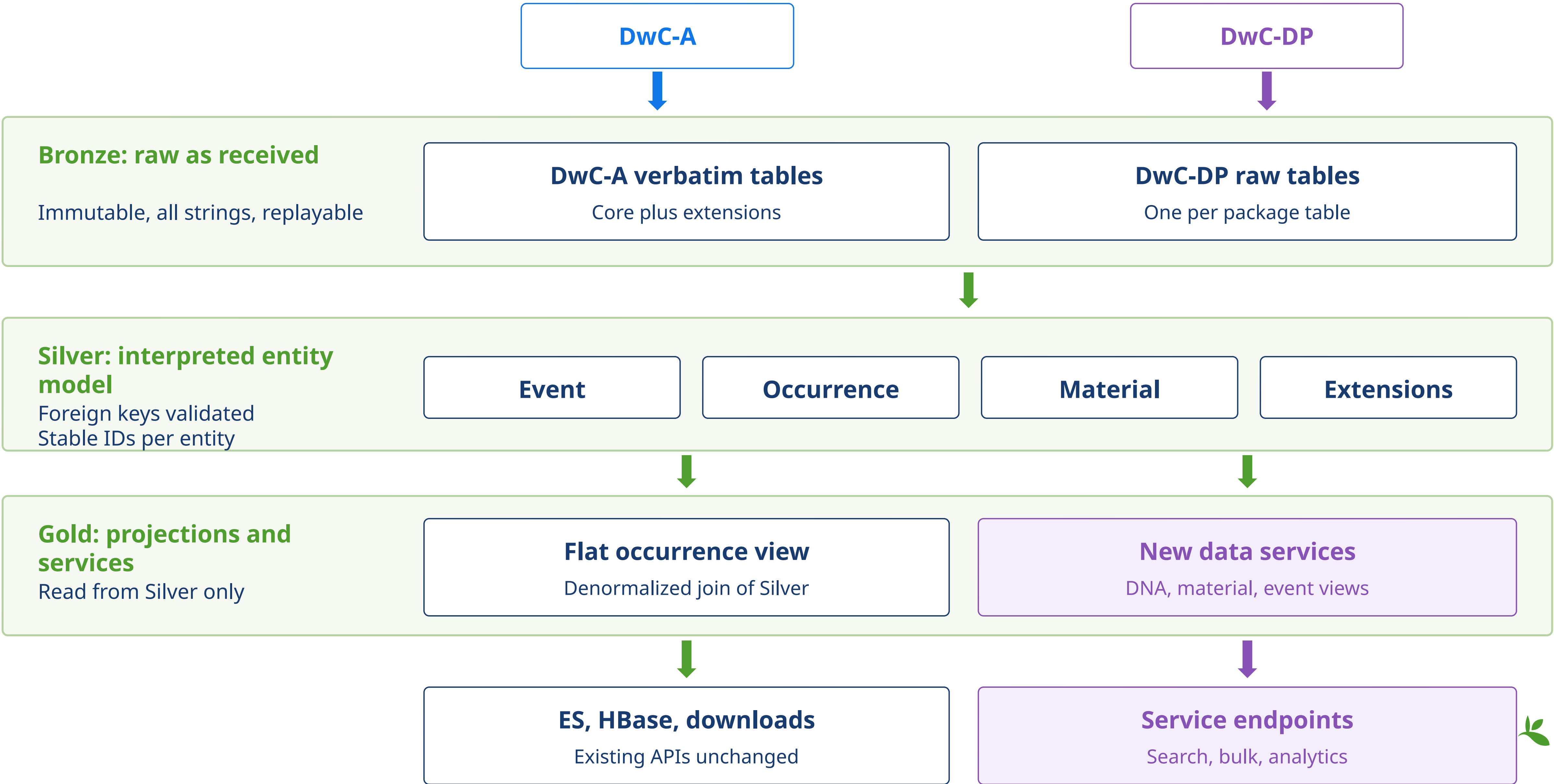
A proposed path from source archives to specialised data services



Illustrative table types and services



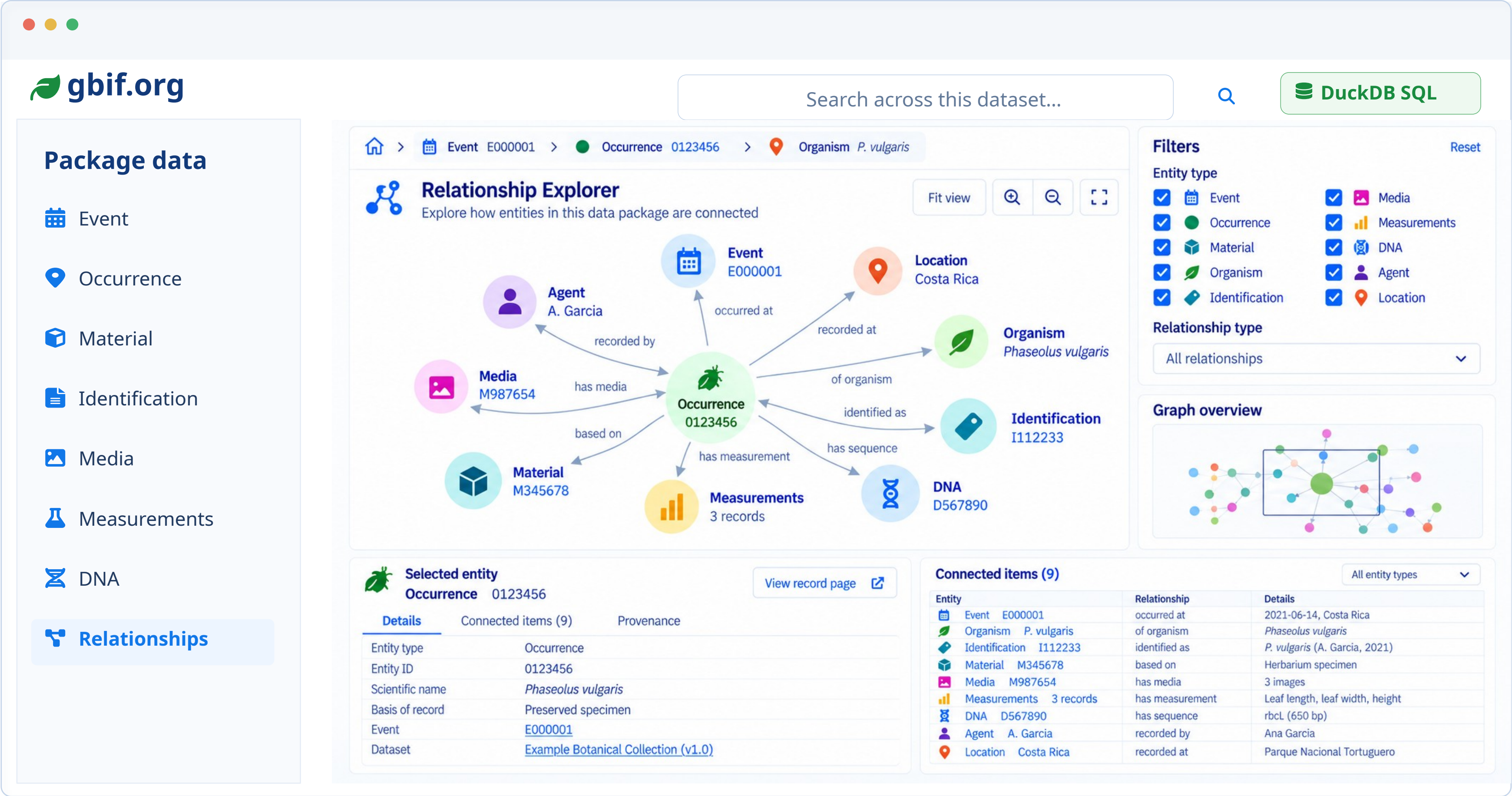
DwC-DP/DwC-A in a Medallion Architecture - Dataflow



Next generation of Dataset views



More interactive, build for Highly Relational Data



Illustrative concept — subject to change



Interactive custom Statistics



Deep insight into dataset



Customizable queriable

Discussion and feedback

Fill the feedback document!

